

MA1521 L1 Week 06 Mon 20260914

(Repeat of a previous question; but this time, use Lagrange multipliers.)

A rectangular produce box is to be made of cardboard; the sides of single thickness, the ends of double thickness, and the bottom of triple thickness, with the top left open. Its volume is to be 1 cubic foot; what should be its proportions in order to use the least cardboard?

A rectangular box is placed in the first octant so that one corner Q is at the origin and the three sides adjacent to Q lie in the coordinate planes. The corner P diagonally opposite Q lies on

a) the plane $x + 2y + 3z = 18$

b) the ellipsoid $x^2 + 2y^2 + 4z^2 = 12$.

Using Lagrange multipliers, tell for which point P the box will have the largest volume, and tell how you know it gives a maximum point.

Using Lagrange multipliers, tell which point P in the first octant and on the surface $x^3y^2z = 6\sqrt{3}$ is closest to the origin.

In an open-top wooden drawer, the two sides and back cost \$2/sq.ft., the bottom \$1/sq.ft. and the front \$4/sq.ft. Using Lagrange multipliers, show that the following problems lead to the same set of three equations in λ , plus a different fourth equation, and they have the same solution.

- a) Find the dimensions of the drawer with largest capacity that can be made for a total wood cost of \$72.
- b) Find the dimensions of the most economical drawer having volume 24 cu. ft.